



TARGET-90

JKBOSE CLASS-11TH

Bhaichara
ON
TOP

REDOX REACTION

LECTURE-1

CHEMISTRY

Vishal Sir



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Topics To Be Covered

1 ...

2 ...

3 ...

4 ...

UNIT-VII: REDOX REACTION

04 MARKS

Concept of oxidation and reduction, redox reactions, oxidation number, balancing of chemical equations in redox reactions, applications of redox reactions.

Concept of Oxidation and Reduction Reactions



1. Classical Concept

(a) Oxidation:

Process in which oxygen or any electronegative element is added

Example: $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$

Process in which hydrogen or any electropositive element is removed

Example: $2\text{H}_2\text{S} + \text{O}_2 \rightarrow 2\text{S} + 2\text{H}_2\text{O}$

(b) Reduction:

Process in which hydrogen or any electropositive element is added

Example: $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$

Process in which oxygen or any electronegative element is removed

Example: $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$

2. Modern Concept

(Electron Transfer Concept)

(a) Oxidation:

Loss of electrons (e^-)

Example: $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + e^-$

Results in increase of positive charge or decrease of negative charge.

(b) Reduction:

Gain of electrons (e^-)

Example: $\text{Fe}^{3+} + e^- \rightarrow \text{Fe}^{2+}$

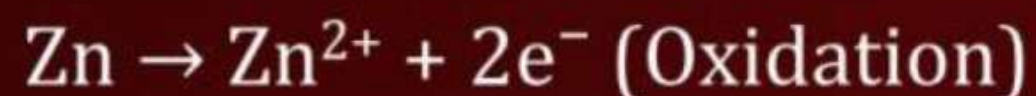
Results in decrease of positive charge or increase of negative charge.

Concept of Oxidation and Reduction Reactions



3. Redox Reaction

Reaction in which oxidation and reduction occur simultaneously.



4. Oxidising and Reducing Agents

Oxidising Agent (Oxidant):

Substance that gains electrons (gets reduced).

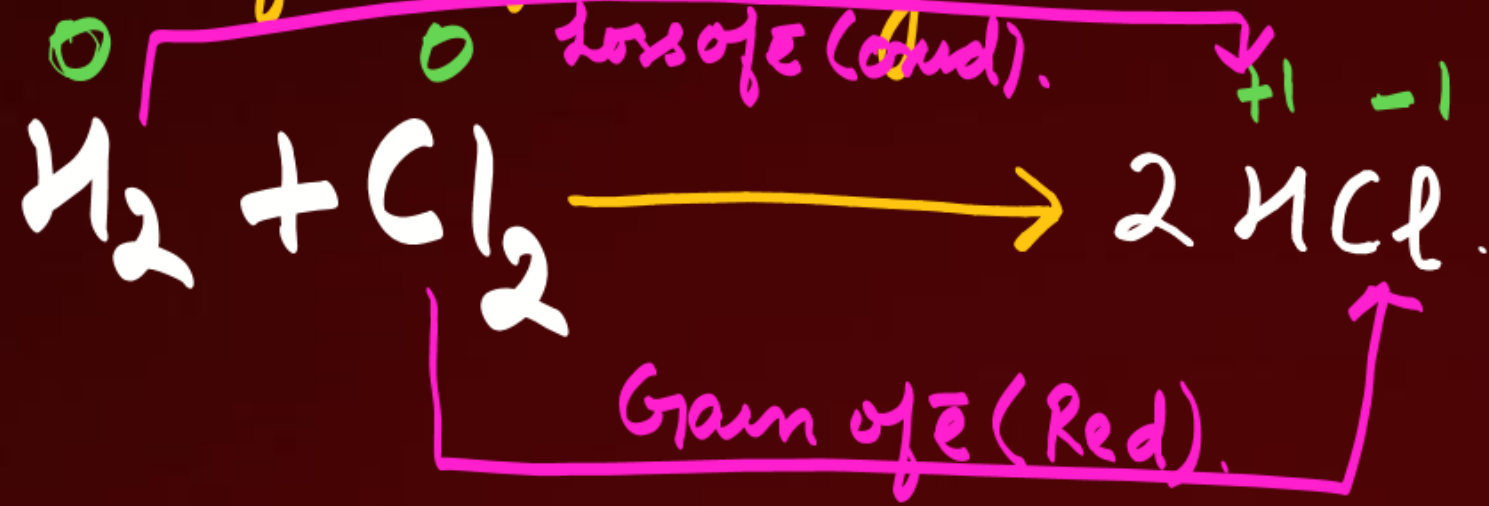
Example: Cu^{2+} act as oxidising agent in above rxn.

Reducing Agent (Reductant):

Substance that loses electrons (gets oxidised).

Example: Zn in $\text{Zn} + \text{CuSO}_4$ reaction.

Which of the following will act as oxidising agent?



↓
Reduction (Gain of e^-)

a) H_2

b) Cl_2

c) HCl

d) None.

Oxidation Number

99% Sawal.

Oxidation State or Number:

- Charge present on any atom in a compound when all the other atoms are removed from that compound.



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Calculation of Oxidation Number



- Free/elementary state:** Oxidation number = 0 (e.g., H_2 , O_2 , O_3 , Fe, C (diamond), Br_2 , P_4 , S_8).
- Monoatomic ion:** Oxidation number = charge on ion
e.g., $\text{K}^+ = +1$, $\text{Ca}^{2+} = +2$, $\text{Al}^{3+} = +3$, $\text{Cl}^- = -1$,
- Metal + Non-metal compounds:** Metal = positive, Non-metal = negative
(e.g., $\text{NaCl} \rightarrow \text{Na} = +1$, $\text{Cl} = -1$).
- Hydrogen:** Usually +1; in metal hydrides (LiH , NaH , CaH_2 etc.) = -1.
- Oxygen:** Usually -2.
 - Peroxides (H_2O_2 , Na_2O_2): -1 each. (O^{-1})
 - Superoxides (KO_2 , RbO_2): $-\frac{1}{2}$ each. ($\text{O}^{-\frac{1}{2}}$)
 - With fluorine (OF_2 : +2; O_2F_2 : +1 each).
- Fluorine:** Always -1 (most electronegative). Other halogens = -1, except when bonded to a more electronegative atom.
- Neutral molecules:** Sum of oxidation numbers = 0 (e.g., NH_3).
- Polyatomic ions:** Sum of oxidation numbers = net charge on ion (e.g., SO_4^{2-}).

Group 1	Group 2
$+1$	$+2$
H^{+1}	
Li^{+1}	Be^{+2}
Na^{+1}	Mg^{+2}
K^{+1}	Ca^{+2}
Rb^{+1}	Sr^{+2}
Cs^{+1}	Ba^{+2}
Fr^{+1}	Ra^{+2}



Total char = -2

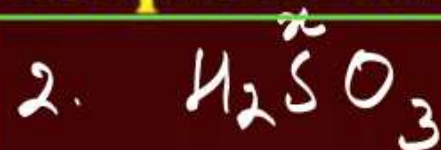
Calculate the oxidation number of sulphur in the following molecules/ions:



$$2(+1) + x = 0$$

$$2 + x = 0$$

$$x = -2$$



$$2(+1) + x + 3(-2) = 0$$

$$2 + x - 6 = 0$$

$$2 - 6 + x = 0$$

$$-4 + x = 0$$

$$x = +4$$



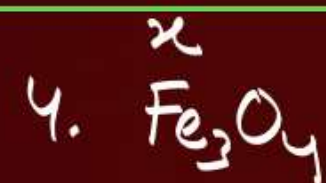
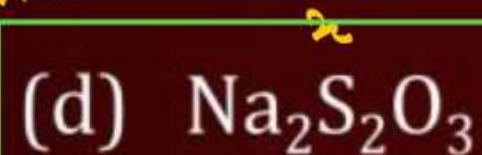
$$x + 4(-2) = -2$$

$$x - 8 = -2$$

$$x = 8 - 2 = +6$$



3. w.



$$3x + 4(-2) = 0$$

$$3x - 8 = 0$$

$$3x = +8$$

$$x = \frac{8}{3}$$



$$1 + x + 4(-2) = 0$$

$$1 + x - 8 = 0$$

$$1 - 8 + x = 0$$

$$-7 + x = 0$$

$$x = +7$$



$$2(+1) + 2x + 7(-2) = 0$$

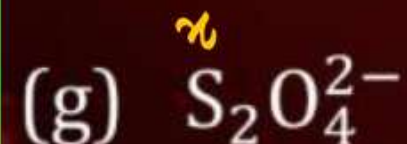
$$2 + 2x - 14 = 0$$

$$2x + 2 - 14 = 0$$

$$2x - 12 = 0$$

$$2x = 12$$

$$x = \frac{12}{2} = +6$$



⑦

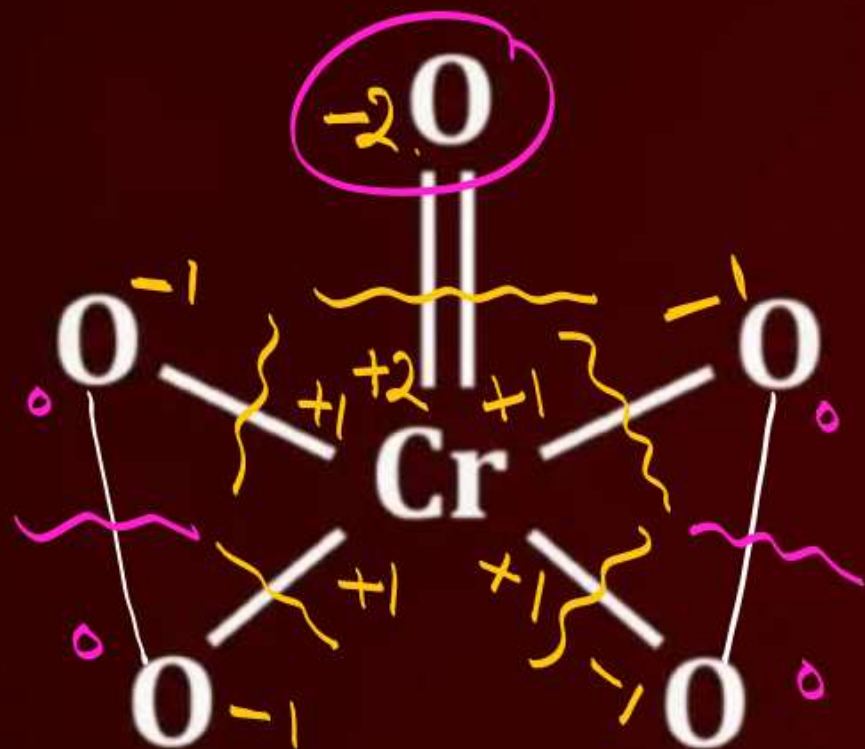
Write formulas for the following compounds:

- (a) Mercury (II) chloride
- (b) Nickel (II) sulphate
- (c) Tin (IV) oxide
- (d) Thallium (I) sulphate
- (e) Iron(III) sulphate
- (f) Chromium(III) oxide

Special Cases



1. CrO_5

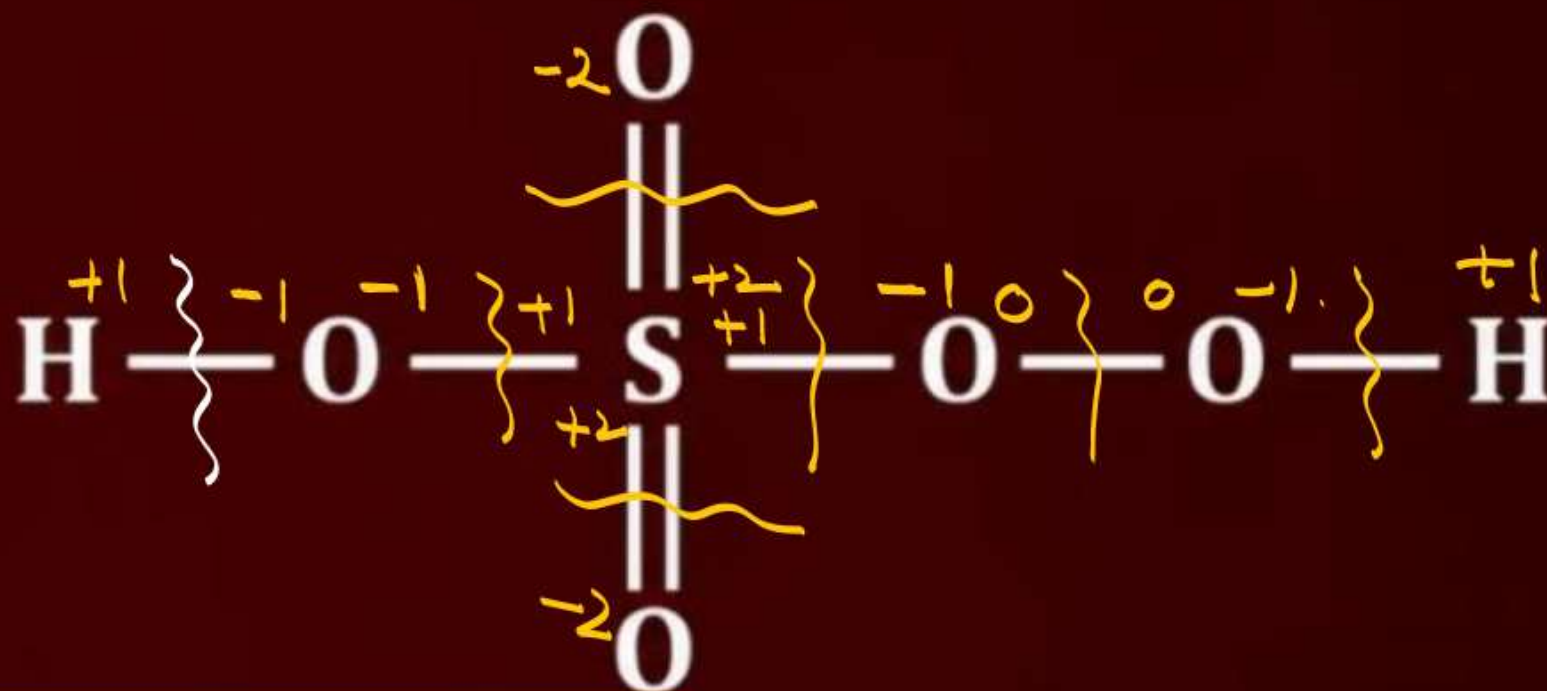


Cr^{+6}

Four O^{-1}

One O^{-2}

2. H_2SO_5 (Caro's acid)



S^{+6}

Two H^{+1}

Three O^{-2}

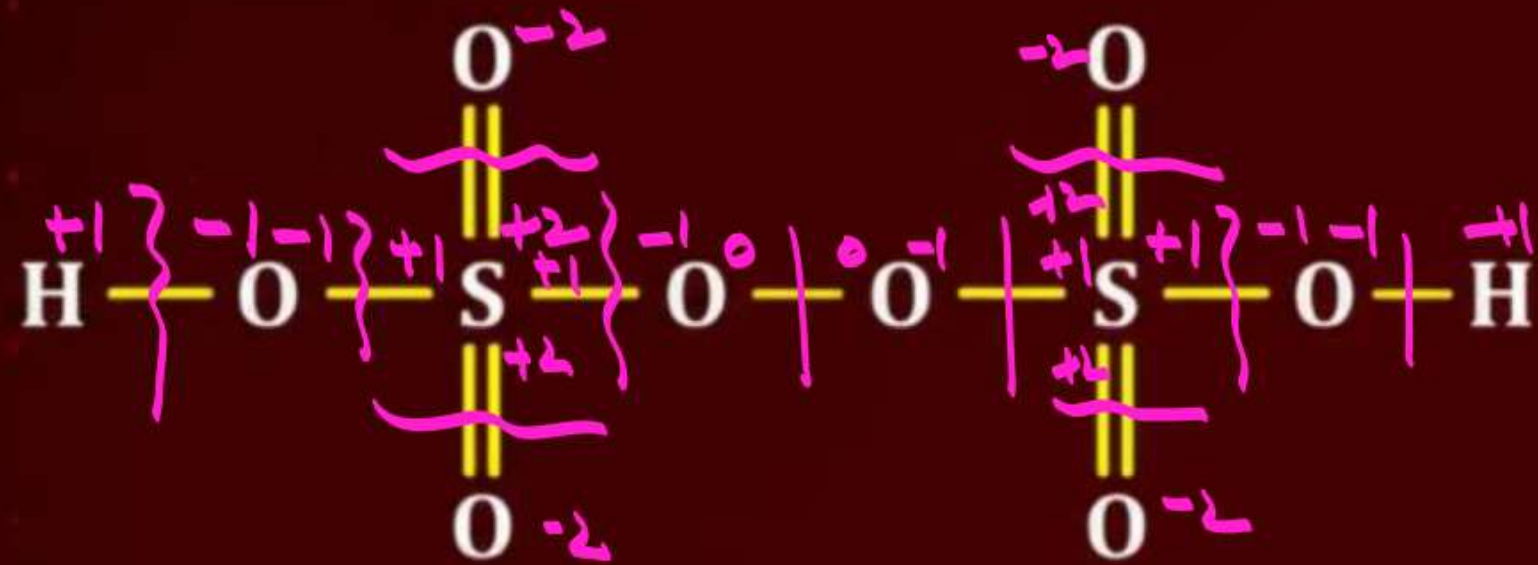
Two O^{-1}

$\text{F} > \text{O} > \text{N}$

Special Cases

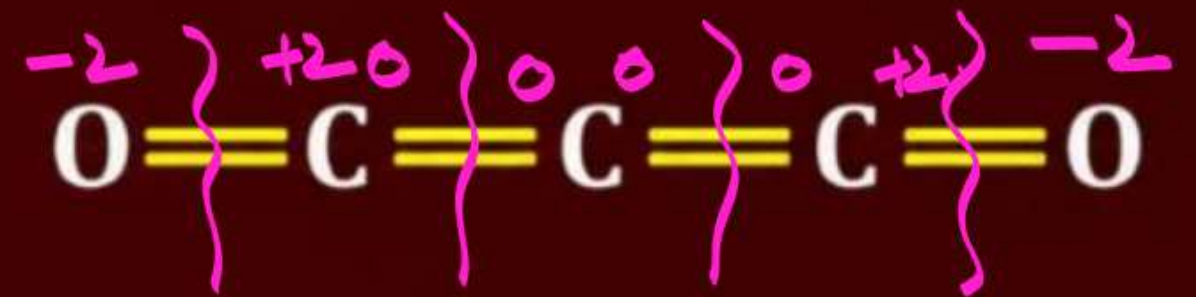


3. $\text{H}_2\text{S}_2\text{O}_8$ (Marshall's acid)



Two S^{+6}
 Two H^{+1}
 Six O^{-2}
 Two O^{-1}

4. C_3O_2 (Carbon suboxide)



Two O^{-2}
 Two C^{+2}
 One C^0

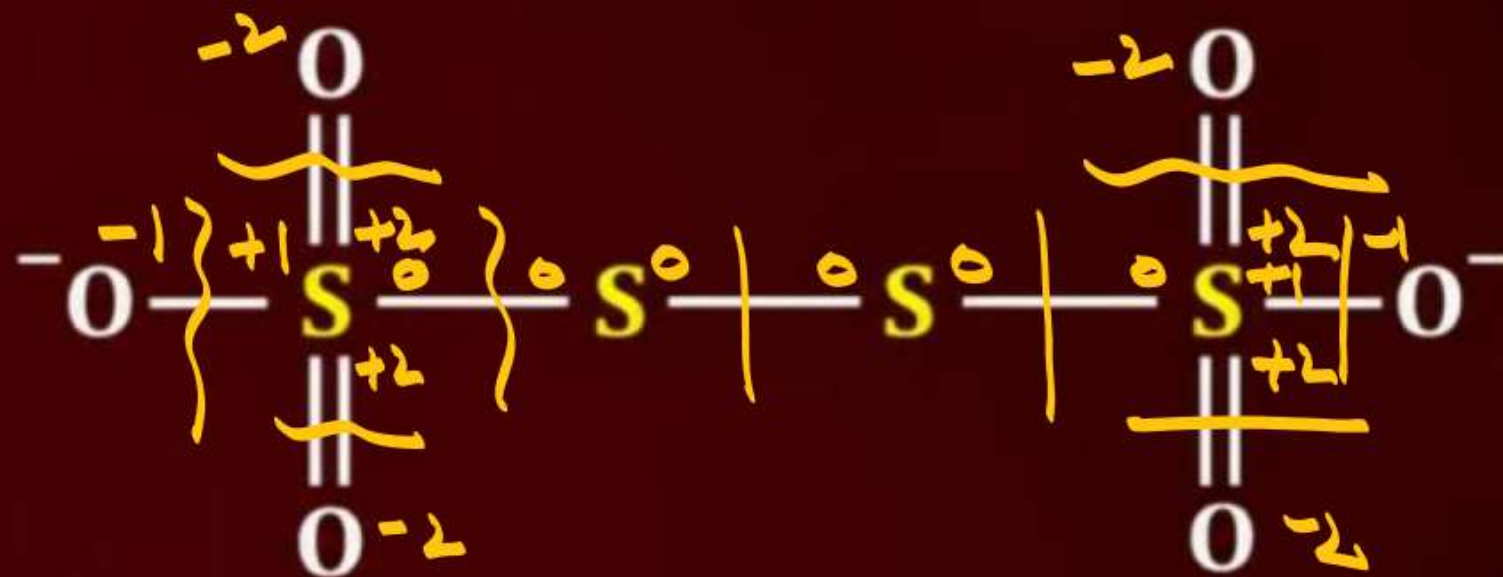
Special Cases



5. Br_3O_8 (Tribromo octaoxide)



6. $\text{H}_2\text{S}_4\text{O}_6$ or $\text{S}_4\text{O}_6^{2-}$ (Tetrathionate)



Two S^{+5}
Six O^{-2}
Two S^0

How to calculate maximum & minimum Oxidation State of Element.

a) Maximum Oxidation state of Element = Valence $\bar{e}$

e.g. Nitrogen = 5 valence $\bar{e}$ = $+5$
 Sulphur = 6 valence $\bar{e}$ = $+6$

b) Minimum Oxidation state of Element = Valence $\bar{e}$ - 8

e.g. Nitrogen = $5 - 8$
 = -3
 Sulphur = $6 - 8$
 = -2

- When element is in its Max. O.S. = Act as Oxidising Agent.
- When element is in its minimum O.S. = acts as Reducing agent.
- When element is in its middle O.S. = can act as Oxidising as well as Red

Q. What is Max & min O.S of P.

a) Max.
-3 Min
+5

b) +5 -3.

c) +8 -6

d) -6 +8.

(15) (N) (P)

↓

Value e = 5e

Max O.N = Value e = +5

Min. ON = Value e - 8 = 5 - 8 = -3

QUESTION



$\overset{+4}{S}$ $\xrightarrow{\text{Max } +6}$
 $\xrightarrow{-2}$
 H_2S acts only as reductant where as SO_2 acts as both oxidant and reductant. Why?

Sol: $\rightarrow$ S in H_2S has -2 Oxidation state i.e its min. O.S so it can only act as reductant. [2025]

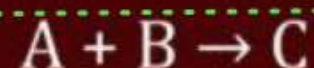
But in SO_2 , S has $+4$ O.N i.e it has middle O.N because S has $+6$ Max. O.S. i.e can act as OA as well as RA.

Types of Redox Reactions

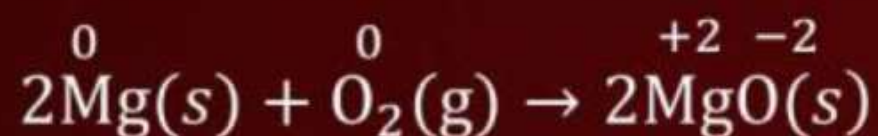
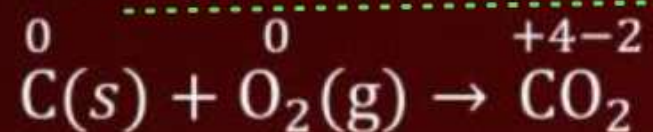


1. Combination reactions

The chemical reactions in which two or more substances (elements or compounds) combine to form a single substance are called combination reactions. A combination reaction may be expressed as:



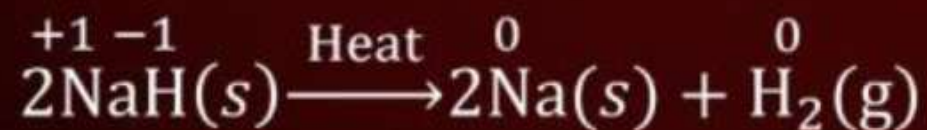
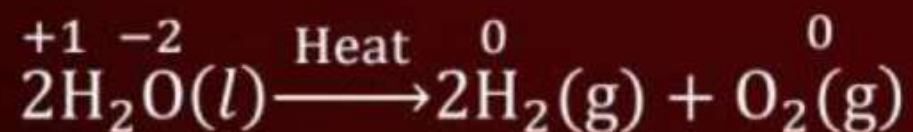
Some common combination redox reactions are:



2. Decomposition reactions

The chemical reactions in which a compound breaks up into two or more simpler substances are called decomposition reactions.

Some common examples of redox decomposition reactions are:



Types of Redox Reactions



3. Displacement reactions

The reactions in which less reactive species in a compound is replaced by more reactive species of other element are called displacement reactions.

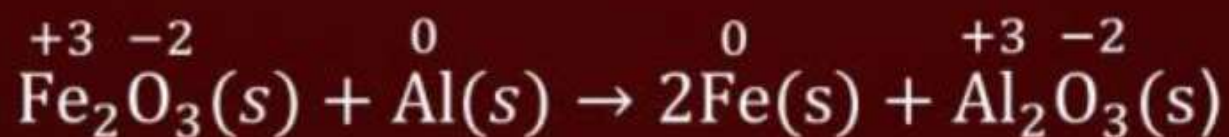
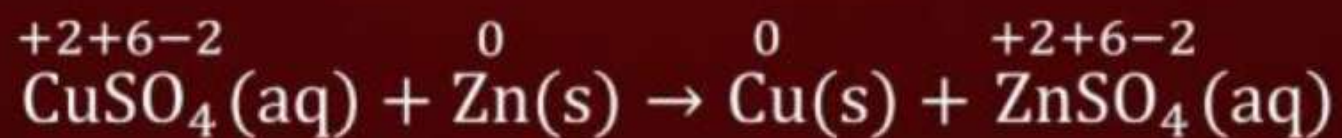
The displacement reactions are of two types:

- (a) Metal displacement and
- (b) Non-metal displacement

(a) Metal displacement reactions.

In these reactions, a metal in a compound can be displaced by another metal in the uncombined state.

Some common examples of metal displacement reactions are:

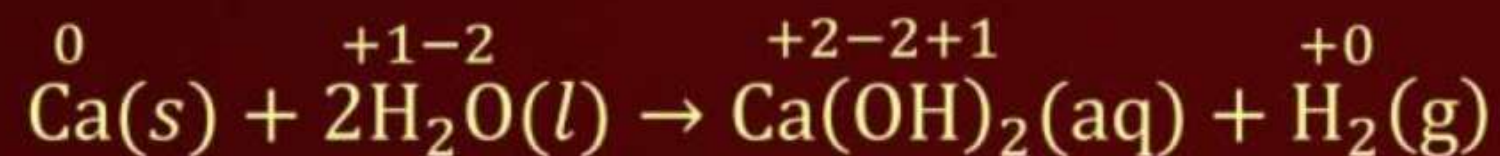
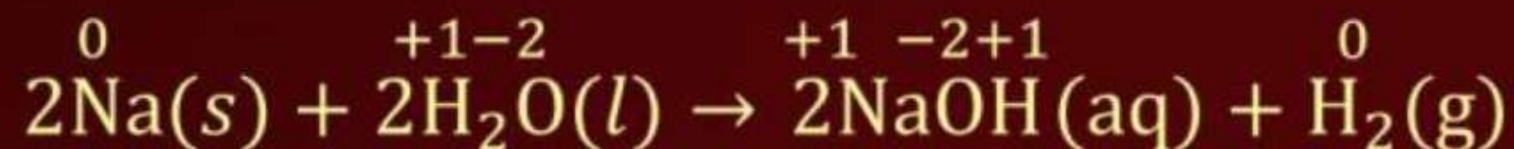


Types of Redox Reactions



(b) Non-Metal displacement reactions.

In these reactions, a metal or a non-metal displaces another non-metal from its compound. In most of these reactions, the non-metal displaced is hydrogen. However, some reactions involving displacement of oxygen or halogens are also known.



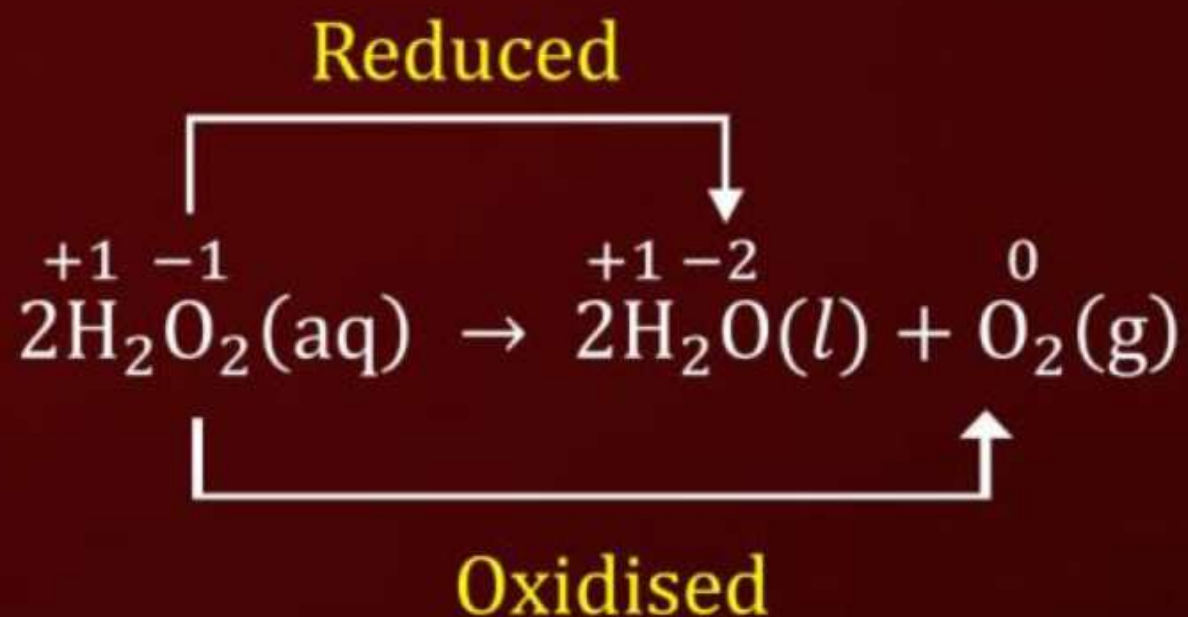
metal $\rightarrow$ Non-metal
NM. $\rightarrow$ in compound

Types of Redox Reactions



4. Disproportionation reactions

A reaction in which the same species is simultaneously oxidised as well as reduced is called a disproportionation reaction. For such redox reactions to occur, the reacting species must contain an element which has at least three oxidation states.



Balancing The Redox Reactions



(A) Oxidation Number Method

The various steps involved in the balancing of redox equations by oxidation number method are:

- Step 1.** Write the skeletal equation of all the reactants and products of the reaction.
- Step 2.** Indicate the oxidation number of each element above its symbol and identify the elements which undergo a change in the oxidation number (O.N.).
- Step 3.** Calculate the increase or decrease in oxidation number per atom and equalise it by multiply with suitable number.
- Step 4.** Balance all atoms other than H and O.

Balancing The Redox Reactions

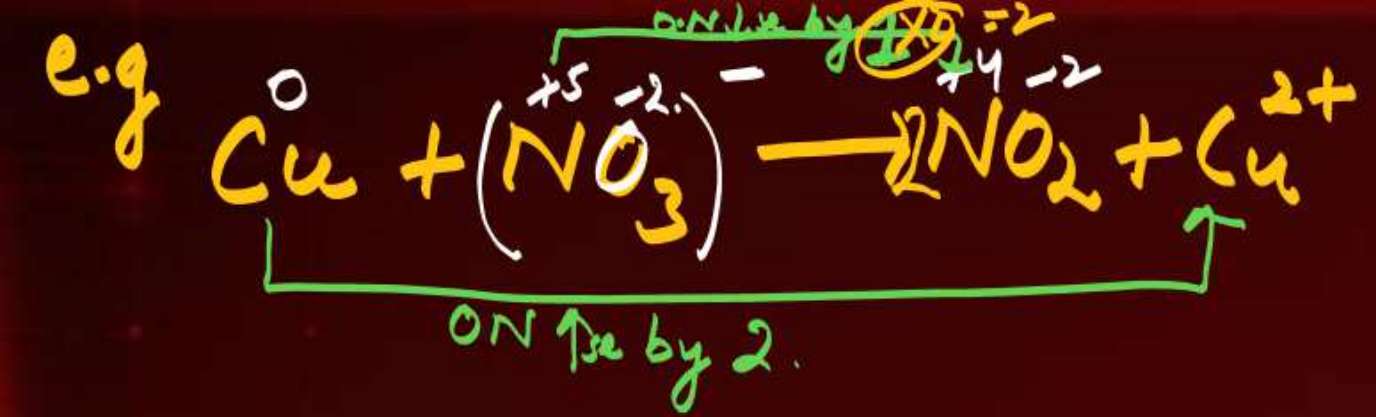


• Balance O by adding H_2O .
• & balance H by adding H^+ .

Step 5. In case

- (a) For acidic medium:** First balance O atoms by adding H_2O molecule to whatever side deficient in O atoms and then balance H atoms by adding H^+ ions to whatever side deficient in H atoms.
- (b) For basic medium:** First balance O atoms by adding H_2O molecules to whatever side deficient in O atoms. The H atoms are then balanced by adding H_2O molecules equal in number to the deficiency of H atoms and an equal number of OH^- ions are added to the opposite side of the equation.

• Balance 'O' by adding H_2O
Balance H by adding $H_2O = H \text{ deficiency}$
then add equal no. of OH^- to opposite side of rxn.

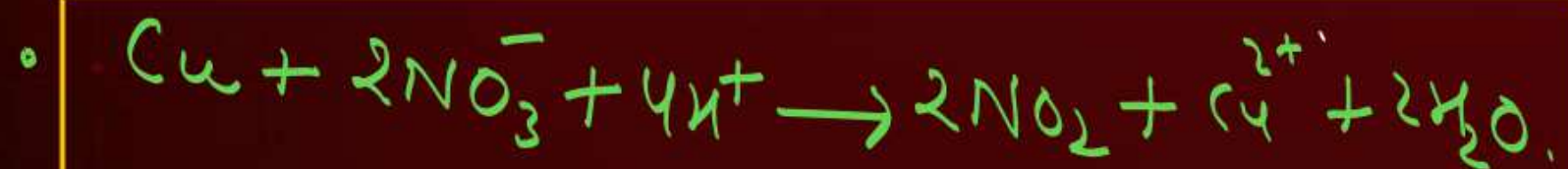


Acidic Medium

Balance the O by H_2O .



Balance the H by H^+ + 2 H_2O



Basic Medium

Balance the O by adding H_2O .



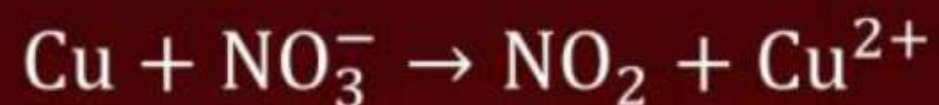
Balance the H atom.



QUESTION



Copper reacts with nitric acid. A brown gas is formed and the solution turns blue. The equation may be written as:



Balance the equation by oxidation number method.

Balancing The Redox Reactions



(B) Ion-Electron Method (Balancing Equations Using Half Reactions)

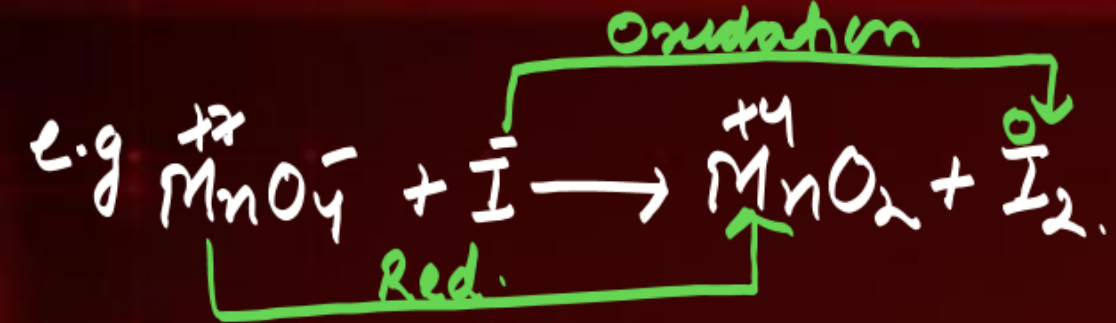
1. Write the ionic form of Chemical Equation.
2. Identify the elements undergoing Oxidation & Reduction.
3. Break the Reaction into two Halves.
 - (i) Oxidation Half
 - (ii) Reduction Half
4. Balance the half reaction
 - (i) Balance all the atoms except 'H' & 'O'.
 - (ii) Balance the Oxygen by adding H_2O Molecule to the Side having less no. of oxygen.
 - (iii) Balance to hydrogen by adding H^+ to the Side having less no. Of Hydrogen.
 - (iv) Balance the charge by adding electron.**
5. Add the two half Such that electron cancel each other.
6. In Basic Medium add as many OH^- ions to the both Side of reaction as H^+ were added previously.

QUESTION



Permanganate (VII) ion, MnO_4^- in basic solution oxidises iodide ion, I^- to produce molecular iodine (I_2) and manganese (IV) oxide (MnO_2). Write a balanced ionic equation to represent this redox reaction.

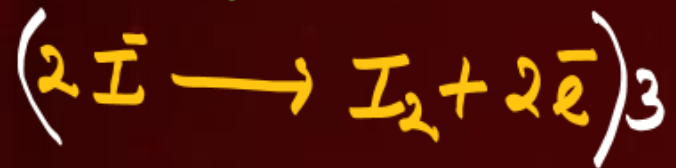




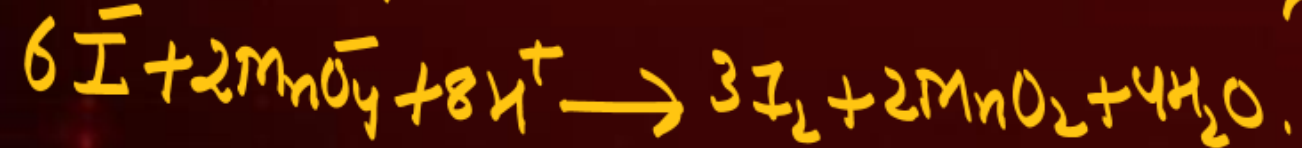
Orud. half



Balancing the charge



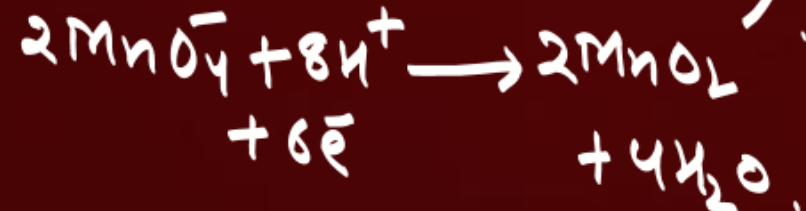
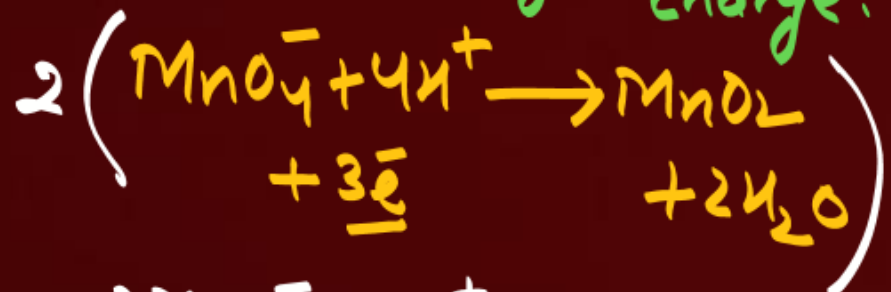
Adding



Red. half



Balancing the charge



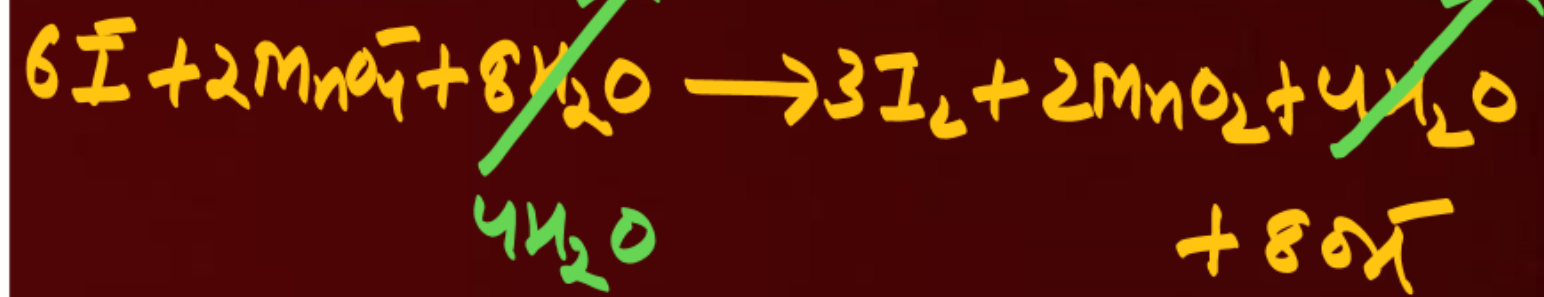
→ Acidic

Basic Med.



8OH⁻

8OH⁻

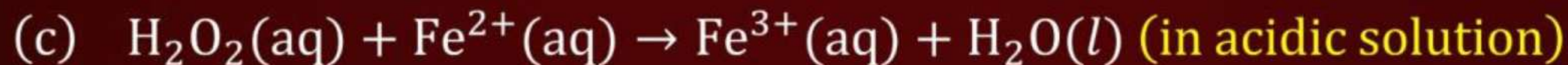


QUESTION

1) O.N method
2) Half Rxn.



Balance the following redox reactions by ion-electron method:



Application of Redox Reactions



1. Photosynthesis

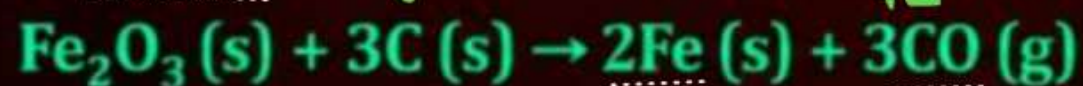
- CO₂ and H₂O are converted to carbohydrates in the presence of sunlight using chlorophyll.
- Here, CO₂ is **reduced** (gain of e⁻) and H₂O is **oxidised** (loss of e⁻).



- C in CO₂: +4 → 0 (**reduction**)
- O in H₂O: -2 → 0 (**oxidation**)

2. In metallurgy

- Metal oxides are reduced to metals using suitable reducing agents like C, CO or H₂.
- Example:** In a blast furnace, Fe₂O₃ is reduced to iron using coke.



- Fe in Fe₂O₃: +3 → 0 (**reduction**)
- C in C: 0 → +2 (**oxidation**)

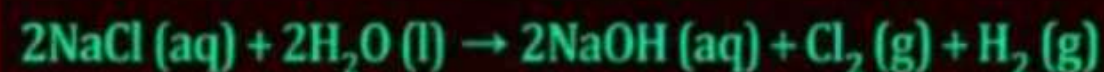


Blast furnace

3. Production of chemicals

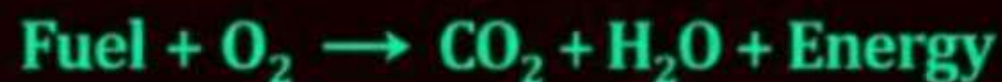
- Many important chemical compounds are prepared by redox reactions, often through electrolysis.
- Examples:** NaOH, Cl₂, F₂, KOH, etc.

Example (Electrolysis of brine):

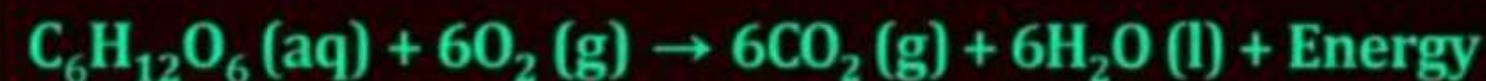


4. Oxidation of fuels

- Fuels are oxidised by O₂ to give CO₂, H₂O and energy.
- Used for cooking, heating, transport and industrial purposes.
- Also used for digestion of food in living organisms.



Example (Glucose):



Application of Redox Reactions



5. Electrochemical cells or batteries.

- Electrochemical cells/batteries are based on **redox reactions** and are widely used as sources of electrical energy. Storage batteries power vehicles such as cars, trucks and buses.



6. Quantitative relationships & stoichiometry.

- Redox titrations are used in quantitative analysis. They involve reactions between oxidising and reducing agents and help determine the amount/concentration of an unknown species.



QUESTION

The oxidation number of oxygen in OF_2 is:

[2025]

1 +2

2 -2

3 +1

4 -1



$$\begin{aligned} & \overset{x}{\text{O}} \text{F}_2 \\ & x + 2(-1) = 0 \\ & x - 2 = 0 \\ & x = +2 \end{aligned}$$

QUESTION



The oxidation number of iron in Fe_3O_4 is:

[2025]

1 +2

2 +3

3 $\frac{8}{3}$

4 $\frac{2}{3}$

QUESTION



An oxidation process involves:

[2025]

- 1 Increase in oxidation number
- 2 Decrease in oxidation number
- 3 Addition of electrons
- 4 Removal of electronegative element

QUESTION



The oxidation number of the two Cl atoms in bleaching powder, CaOCl_2 are:

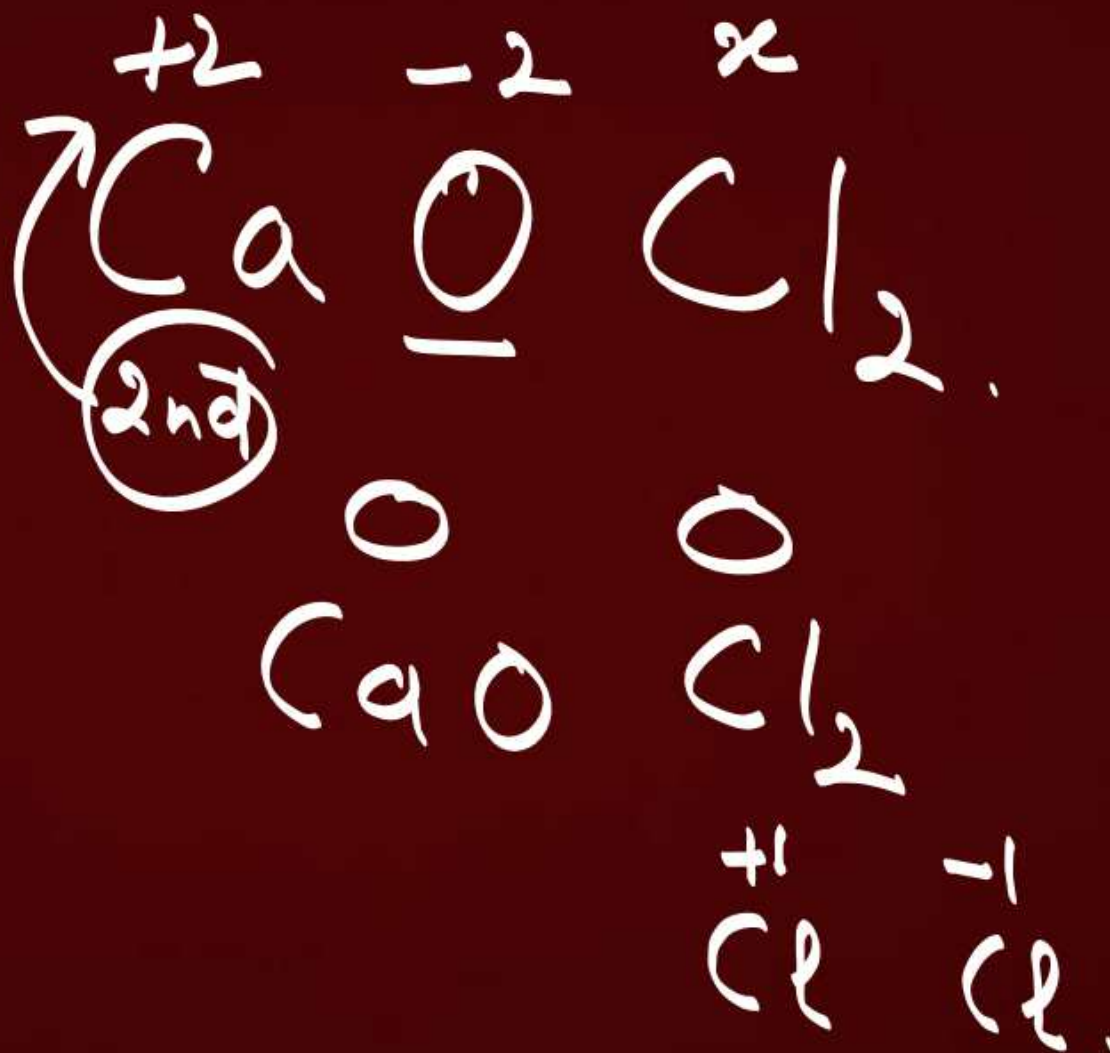
[2024]

1 $-1, -1$

2 $+1, -1$

3 $+1, +1$

4 $0, -1$

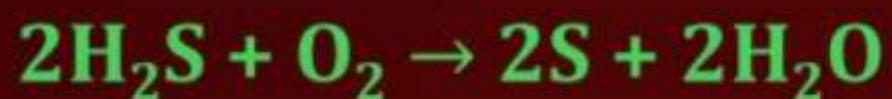


QUESTION



- (a) Define oxidation and reduction on the basis of electron transfer concept. Give example in each case.
- (b) In the reactions given below, identify the species undergoing oxidation and reduction:

[2024]



QUESTION



The oxidation state of Cr in CrO_5 is:

[2024]

1 +10

2 -10

3 +6

4 -6

HOMEWORK



SUMMARY



Thank
You

BOT
♡
BYEE

- ① Dr. Mohd. Areeb
- ② Jamsheed
- ③ Safiya Hassan